

Taller de Ingeniería Informática - Laboratorio N2

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```

root@debian:~# cd /root/practico2
root@debian:~/practico2# docker compose build
[+] Building 4.1s (10/10) FINISHED
=> [internal] load local bake definitions
=> => reading from stdin 482B
=> [internal] load build definition from Dockerfile
=> => transferring Dockerfile: 138B
=> [internal] load metadata for docker.io/library/php:8.2-apache
=> [internal] load Dockerignore
=> => transferring context: 4B
=> [1/3] FROM docker.io/library/php:8.2-apache@sha256:affce43fbd9ac996394671d162726fca9e40ea400a3b94f925b6130852dd
=> => resolve docker.io/library/php:8.2-apache@sha256:affce43fbd9ac996394671d162726fca9e40ea400a3b94f925b6130852dd
=> [internal] load build context
=> => transferring context: 7.3kB
=> Docker 12/31 RUN docker-php-ext-install pdo pdo_mysql
=> [3/3] COPY -- /var/www/html/
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:468026c6df9c149e57f52f1ab9cf9b64b75c9c18d21a9f5ce91b913cebd13
=> => exporting config sha256:e9cf6414ab7c515e3d29ca3d12ed325b071baef4e708e368f3fe3e2a1be4f
=> => exporting attestation manifest sha256:f94eadd1ef2eb0a0fc36d4b00a0f076f913c831403bcb0932d489853e88b74
=> => exporting manifest list sha256:5935b7595b67295b0eedf050979abccc0413585b0b5d424cc8777db7042f019
=> => naming to docker.io/library/practico2-web:latest
=> => unpadding to docker.io/library/practico2-web:latest
=> resolving provenance for metadata file
[+] build 1/1
< Image practico2-web Built
root@debian:~/practico2#

```

Figure 1: Ejecución de docker-compose build

1 Introducción

Este informe documenta el despliegue de una aplicación web dinámica con operaciones CRUD utilizando contenedores Docker, orquestados con docker-compose, una base de datos MySQL y persistencia mediante volúmenes.

2 Desarrollo

2.1 Estructura de la aplicación

Se implementó una aplicación en PHP con MySQL que permite Crear, Leer, Actualizar y Eliminar registros de una tabla "items". La arquitectura se compone de dos contenedores:

- **web:** Contenedor PHP con Apache, que expone el puerto 8080.
- **db:** Contenedor MySQL 8.0 con persistencia en volumen.

Ambos se comunican a través de una red Docker interna llamada `crud_network`.

2.2 Evidencias de ejecución

3 Conclusión

Se logró desplegar exitosamente una aplicación web CRUD utilizando Docker, cumpliendo todos los requisitos: contenedores separados, volumen persis-

```
root@debian:/reacticore# docker-compose up -d
[+] Up 2/2
+ Container mysql_crud_db Running
+ Container php_crud_app Started
root@debian:/reacticore# docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                               NAMES
c30b7aaa3012   practico2-web   "docker-entrypoint.s..." 54 seconds ago Up 54 seconds 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   php_crud_app
076af0c5c008   mysql:8.0      "docker-entrypoint.s..." 51 minutes ago Up 2 minutes  0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp, 33060/tcp   mysql_crud_db
da5552835b64   odoo:18        "/entrypoint.sh odoo"    31 hours ago   Up 2 minutes  0.0.0.0:8069->8069/tcp, [::]:8069->8069/tcp, 8071-8072/tcp   trabajo-odoo
09-0d00-1
7f089b5d1de   postgres:15    "docker-entrypoint.s..." 31 hours ago   Up 2 minutes  5432/tcp   trabajo-postgres
00-db-1
root@debian:/reacticore#
```

Figure 2: docker-compose up -d y docker ps

```
2026-06-03 20:12:38+00:00 [Note] [Entrypoint]: Stopping temporary server
2026-06-03T20:12:38.482643Z 14 [System] [MY-013172] [Server] Received SHUTDOWN from user root. Shutting down mysqld (version: 8.0.46).
2026-06-03T20:12:38.265355Z 0 [System] [MY-019301] [Server] /usr/sbin/mysqld: Shutdown complete (mysqld 8.0.46)  MySQL Community Server - GPL.
2026-06-03 20:12:38+00:00 [Note] [Entrypoint]: Temporary server stopped

2026-06-03 20:12:38+00:00 [Note] [Entrypoint]: MySQL Init process done. Ready for start up.

2026-06-03T20:12:38.759015Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future release. Please use '--skip-host-cache=0' instead.
2026-06-03T20:12:38.760475Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.0.46) starting as process 1
2026-06-03T20:12:38.773655Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
2026-06-03T20:12:39.515735Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
2026-06-03T20:12:39.896282Z 0 [Warning] [MY-010669] [Server] CA certificate ca.pem is self signed.
2026-06-03T20:12:39.902783Z 0 [System] [MY-013562] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this channel.
2026-06-03T20:12:39.916046Z 0 [Warning] [MY-011819] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld/' in the path is accessible to all OS users. Consider choosing a different directory.
2026-06-03T20:12:39.940350Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var/run/mysqld/mysqlx.sock
2026-06-03T20:12:39.956522Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.0.46' socket: '/var/run/mysqld/mysqld.sock' port: 3306 MySQL Community Server - GPL.
2026-06-03 21:00:47+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.46-1.el9 started.
2026-06-03 21:00:48+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
2026-06-03 21:00:48+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.46-1.el9 started.
2026-06-03T21:00:48.694703Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future release. Please use '--skip-host-cache=0' instead.
2026-06-03T21:00:48.726942Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.0.46) starting as process 1
2026-06-03T21:00:48.750175Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
2026-06-03T21:00:50.487706Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
2026-06-03T21:00:52.392754Z 0 [System] [MY-010223] [Server] Starting XA crash recovery...
2026-06-03T21:00:53.495444Z 0 [System] [MY-010223] [Server] XA crash recovery finished.
2026-06-03T21:00:53.417342Z 0 [Warning] [MY-010669] [Server] CA certificate ca.pem is self signed.
2026-06-03T21:00:53.417515Z 0 [System] [MY-013562] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this channel.
2026-06-03T21:00:53.437322Z 0 [Warning] [MY-011819] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld/' in the path is accessible to all OS users. Consider choosing a different directory.
2026-06-03T21:00:53.556592Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var/run/mysqld/mysqlx.sock
2026-06-03T21:00:53.567380Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.0.46' socket: '/var/run/mysqld/mysqld.sock' port: 3306 MySQL Community Server - GPL.
root@debian:/reacticore#
```

Figure 3: Logs del contenedor MySQL mostrando inicio correcto

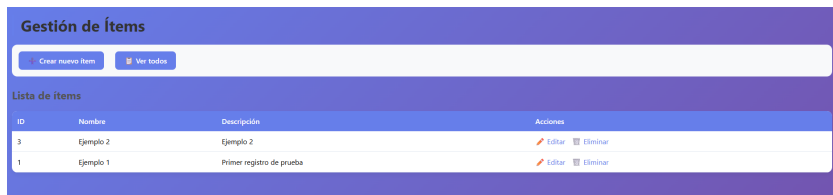


Figure 4: Aplicación corriendo en http://localhost:8080 con operaciones CRUD funcionales

tente, red virtual, y documentación completa. El sistema es reproducible y portable.

4 Anexos

- Repositorio GitHub: <https://github.com/Vicho255/Practico-2.git>